Computer Networks: Week-3 Assignment

1. Which protocol is primarily used to transfer web pages between a client and a web server?
   A. DNS
   B. FTP
   C. HTTP
   D. SMTP

   Ans. C (HTTP)

2. What is the primary function of DNS?
   A. Transfer files between computers
   B. Encrypt network traffic
   C. Manage email delivery
   D. Convert domain names into IP addresses

   Ans. D (Convert domain names into IP addresses)

3. Which of the following is an example of a text-based protocol?
   A. HTTP/2
   B. DNS
   C. HTTP/1.1
   D. MQTT

   Ans. C (HTTP/1.1)

4. Which of the following is a binary protocol?
   A. HTTP/2
   B. DHCP
   C. HTTP/1.1
   D. SMTP

   Ans. A (HTTP/2)

5. What is a non-persistent HTTP connection?
   A. A single TCP connection is used for multiple requests
   B. A new TCP connection is established for each HTTP request/response
   C. No TCP connection is required
   D. The connection remains open indefinitely

   Ans. B (A new TCP connection is established for each HTTP request/response)

6. A client establishes a connection, requests the homepage, receives it, and then requests the logo image on the same connection. The server keeps the connection open for future requests. What is this HTTP behavior called?
   A. Persistent HTTP
   B. Non-persistent HTTP
   C. HTTP with connection close
   D. Stateless HTTP

   Ans. A (Persistent HTTP)

7. Which HTTP method is mainly used to retrieve data from a server?
   A. POST
   B. PUT
   C. GET
   D. DELETE

   Ans. C (GET)

8. Which character encoding is commonly used by text-based protocols?
   A. Unicode
   B. Binary Coded Decimal
   C. EBCDIC
   D. ASCII or UTF-8

   Ans. D (ASCII or UTF-8)

9. A user submits a form on a website, but the server cannot understand the request because the data format is incorrect. Which status code will the server return?
   A. 200
   B. 301
   C. 400
   D. 404

   Ans. C (400)

10. In non-persistent HTTP, how many TCP connections are required to fetch a web page with 10 images?
    A. 1
    B. 10
    C. 11
    D. 12

    Ans. C (11)